

Generation Interconnection System Impact Study Report

For Queue Project Q21-1912

Bear Island 138 kV
September 21, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1912. The Interconnection Customer proposes a 132.2 MW solar photovoltaic facility interconnecting at the Bear Island 138 kV substation in the Virginia area. The request was received on October 18, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 684 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$25,618,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
EL DORADO REFINERY-LOOKOUT 138 KV	BEAR ISLAND-DUNN SIDING 345 KV	133.7	316	6.4
BEAR ISLAND 138 kV/1 kV Transformer Bank	CHANCELLOR ROW-BEAR ISLAND 345 KV	115.1	624	41.8
LOOKOUT-NATHAN 138 KV	BEAR ISLAND 345-1 kV Auto	114.4	405	5.3
MORRELL-EL DORADO REFINERY 138 KV Line	DUNN SIDING SUBSTATION-BEAR ISLAND 345 KV line	114.2	334	6.4
NORTH DOSWELL-TRENTON DIESEL 138 KV	DUNN SIDING-BEAR ISLAND 345 KV line	111.2	272	17.4
TRENTON IESEL-LAPORTE JCT 138 KV	BEAR ISLAND SUBSTATION-DUNN SIDING 345 KV	111.2	278	17.7
LAPORTE JUNCTION-BEAR ISLAND 138 KV	DUNN SIDING-BEAR ISLAND 345 KV line	110.7	292	17.7
N DOSWELL-BEAR ISLAND 138 KV	BEAR ISLAND-DUNN SIDING 345 KV L	110.6	291	17.4
FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV L	BIRCHWOOD-LEAVITT 345 KV line	108.2	3,520	5.7
NORTH ANNA SUBSTATION-DUNN SIDING 345 KV	FREDERICKSBURG-DUNN SIDING 345 KV	106.3	1,300	22.2
CITY WEST DIESEL PLANT SUBSTATION-BRANDYWINE POWER FACILITY 345 KV	BRANDYWINE POWER FACILITY-ROBERTS SW STA 345 KV L	105.5	1,321	6.4
CAL SIERRA DEVELOPMENT-BRANDYWINE POWER FACILITY 345 KV	ROBERTS SW STA-GLENANNAN 345 KV line	104.8	1,457	6.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
POLK CITY-NESTUCA 345 KV L	BIRCHWOOD-NAUVOO 345 KV	104.6	1,802	5.2
NAUVOO-CITY W DIESEL PLANT 345 KV line	NAUVOO-ROBERTS SW STA 345 KV	103.8	2,204	8.9
MORGANTOWN-SHINGLETOWN 345 KV	ROBERTS SW STA-NAUVOO 345 KV	103.7	1,139	5.4
BIRCHWOOD-DIST 1000 138 KV	LEAVITT-BIRCHWOOD 345 KV	103.0	789	13.6
TRENTON DIESEL-DST 1000 138 KV	BIRCHWOOD-LEAVITT 345 KV	102.7	804	13.6
POLK CITY-FREDERICKSBURG 345 KV Ckt 1	FREDERICKSBURG-POLK CITY 345 KV Cir. 2	101.8	1,706	7.5
FREDERICKSBURG SUBSTATION-POLK CITY 345 KV #2	FREDERICKSBURG-POLK CITY 345 KV	101.8	1,706	7.5
ROBERTS SW STA-NAUVOO 345 KV	CITY W DIESEL PLANT-NAUVOO 345 KV Line	101.8	2,377	9.5
MONROE NE SUBSTATION-CEDAR CREEK 345 KV	CO LINE RD-MONROE NE 500 KV	101.3	1,587	8.0
GLENANNAN-ROBERTS SW STA 345 KV L	CAL SIERRA DEVELOPMENT-GLENANNAN 345 KV L	100.9	1,974	9.7
FREDERICKSBURG-DUNN SIDING 345 KV	NAUVOO-BIRCHWOOD 345 KV	100.3	2,368	20.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EL DORADO REFINERY-LOOKOUT 138 KV	Rebuild line with higher-capacity conductor	\$39,062,000	10% (\$3,960,000)
BEAR ISLAND 138 kv/1 kv Transformer Bank	Replace transformer with higher-rated unit	\$6,032,000	6% (\$384,000)
NATHAN-LOOKOUT 138 KV Line	Rebuild line with higher-capacity conductor	\$14,404,000	7% (\$972,000)
MORRELL-EL DORADO REFINERY 138 KV Line	Rebuild line with higher-capacity conductor	\$5,513,000	8% (\$454,000)
TRENTON DIESEL-NORTH DOSWELL 138 KV	Rebuild line with higher-capacity conductor	\$2,231,000	81% (\$1,813,000)
TRENTON DIESEL-LAPORTE JCT 138 KV	Rebuild line with higher-capacity conductor	\$3,349,000	81% (\$2,721,000)
LAPORTE JCT-BEAR ISLAND 138 KV line	Rebuild line with higher-capacity conductor	\$9,393,000	81% (\$7,632,000)
N DOSWELL-BEAR ISLAND 138 KV	Rebuild line with higher-capacity conductor	\$4,924,000	81% (\$4,001,000)
FLUVANNA P.S-LOUISA GENERATION FACILITY 345 KV L	Reconductor line	\$22,299,000	0% (\$63,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NORTH ANNA SUBSTATION-DUNN SIDING 345 KV	Reconductor line	\$9,794,000	1% (\$142,000)
BRANDYWINE POWER FACILITY-CITY W DIESEL PLANT 345 KV	Reconductor line	\$7,245,000	1% (\$43,000)
CAL SIERRA DEVELOPMENT-BRANDYWINE POWER FACILITY 345 KV	Reconductor line	\$12,035,000	1% (\$84,000)
NESTUCA-POLK CITY 345 KV	Reconductor line	\$14,243,000	1% (\$131,000)
NAUVOO-CITY W DIESEL PLANT 345 KV line	Reconductor line	\$17,239,000	1% (\$142,000)
MORGANTOWN-SHINGLETOWN 345 KV Line	Reconductor line	\$25,045,000	4% (\$1,024,000)
BIRCHWOOD-DIST 1000 138 KV line	Reconductor line	\$11,259,000	5% (\$515,000)
TRENTON DIESEL-DIST 1000 138 KV	Reconductor line	\$9,529,000	5% (\$436,000)
FREDERICKSBURG-POLK CITY 345 KV Circuit 1	Reconductor line	\$28,442,000	1% (\$256,000)
FREDERICKSBURG SUBSTATION-POLK CITY 345 KV #2	Reconductor line	\$31,695,000	1% (\$285,000)
NAUVOO-ROBERTS SW STA 345 KV L	Reconductor line	\$20,547,000	1% (\$161,000)
CDAR CRK-MONROE NE 345 KV	Reconductor line	\$8,836,000	1% (\$49,000)
GLENANNAN-ROBERTS SW STA SUBSTATION 345 KV line	Reconductor line	\$23,352,000	1% (\$163,000)
FREDERICKSBURG-DUNN SIDING 345 KV	Reconductor line	\$15,700,000	1% (\$187,000)
	Total allocated to project		\$25,618,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.